EEG-based Emotion Detection Using Unsupervised Transfer Learning

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***Abstract*— Emotion classification using EEG signal processing has the potential of significantly improving the social integration of patients suffering from neurological disorders such as Amyotrophic Lateral Sclerosis (ALS) or the acute stages of Alzheimer's disease. One important challenge to the implementation of high-fidelity emotion recognition systems is the inadequacy of EEG data in terms of Signal-to-noise ratio (SNR), duration, and subject-to-subject variability. In this paper, we present a novel, integrated framework for semi-generic emotion detection using (1) independent component analysis for EEG preprocessing, (2) EEG subject clustering by unsupervised learning, and (3) a convolutional neural network (CNN) for EEG-based emotion recognition. The training and testing data was built using the combination of two publicly available repositories (DEAP and DREAMER), and a local dataset collected at Khalifa University using the standard International Affective Picture System (IAPS). The CNN classifier with the proposed transfer learning approach achieves an average accuracy of 70.26% for valence and 72.42% for arousal, which are superior to the reported accuracies of all generic (subject-independent) emotion classifiers.**

I. Introduction

Emotion classifiers based on EEG features, which are potentially suitable for locked-in patients with ALS or Alzheimer, have historically used two overall approaches for training: (1) Subject-dependent—in which the data used for training is collected and tested only for the same participant [1][2], and (2) Subject-independent—in which the training uses data from all volunteers, to be tested in any subject [3][4]. The subject-dependent classifiers are better to distinguish emotions, but they lack a sufficient number of samples. Long sessions with a high number of stimuli on the same subject increase the risk of mental fatigue affecting the EEG readings [5]. One potential solution is to have multiple sessions per subject to increase the number of samples. Unfortunately, multiple sessions suffer from day-to-day variability for the same subject and from the uncertainties on the placement of EEG electrodes. A recent approach to generate artificial EEG data using Generative Adversarial Networks (GAN) aims to solve this problem through data augmentation [6]. However, there are still challenges involved in defining quantitative metrics to assess the artificial EEG frames.

Another solution to the problem of sample scarcity is to adopt subject-independent classifiers using data from multiple subjects, which has typically resulted in poor classification accuracy due to under-fitting. In this paper, we address this problem and propose a semi-independent approach based on unsupervised transfer learning and the extraction of similarities among the EEG frequency patterns of the various subjects. To the best of our knowledge, there are no publication in the open literature on using transfer learning in the context of EEG emotion detection. C. S. Wei *et. al.* [7] used inter-subjects transfer learning to enhance an EEG-based drowsiness detector. The closest contribution to our technique is the transfer recursive feature elimination (T-RFE) proposed by Y. Zhong *et. al.* [8]. It is motivated by considerations similar to ours but their approach extracts common EEG features across all the subjects, while ours is based on clustering similar subjects within the training set.

This paper is organized as follows. Section II presents the local EEG data collection and the preprocessing steps. Section III describes the CNN classifier. Section IV describes the transfer learning technique used while Section V illustrates the semi-independent framework and data flow. Section VI presents the results, and Section VII concludes the paper.

II. Experimental Framework

This study combines two state-of-the-art datasets for emotion detection, DEAP [1] and DREAMER [3], with an in-house dataset collected using a low-cost off-the-shelf device, Emotiv Epoc Plus. In contrast to DEAP and DREAMER, the goal of the in-house experiments is to increase the number of samples per subject, by using short experiences evoked with the International Affective Picture System (IAPS) stimuli.

A. Local Dataset

In our experiments, each of the 5 subjects ($slocxx$) observes and classifies 3 sets of 60 pictures taken from the IAPS dataset while wearing a commercial EEG cap. The automated protocol starts with a RELAX picture lasting for 5 seconds. Next, it displays, for 6 seconds, a picture to be classified. Last, it shows a rating picture called SAM (Self-Assessment Mankini) [9] that contains the three dimensional model used to describe the emotions (Valence, Arousal and Dominance) [10]. Each of the 60 pictures under classification has its own RELAX and SAM pictures.

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The experimental procedures involving human subjects described in this paper were approved by the Research Ethics Committee.

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B. Datasets Combination

To achieve a universal dataset with a high number of subjects, the datasets DEAP, DREAMER and ours are standardized into one. In contrast to our experiments, the 32 subjects ($sdeaxx$) from DEAP rated the Valence and Arousal of 40 one-minute length videos while wearing a 32-channel (10-20 System) EEG cap sampling at 512 Hz. The 23 subjects ($sdrexx$) from DREAMER, classified 18 videos with variable length (65-393 s) and using the same 14-channel (10-20 System) EEG cap from our study at a sampling rate of 128 Hz. The merger of these three datasets covers a total of 60 subjects and constitutes the largest EEG frequency data repository for emotion detection with 2194 samples for training and testing. Each sample is labeled into four classes (High and Low Arousal/Valence) using the collected manual ratings.

1) Frequency Emotion eXpression Dataset (FEXD): Several prior publications have given enough evidence of the intrinsic correlation between frequency components and brain emotions activity [1][11]. Our Frequency Emotion eXpression Dataset (FEXD) adopts the frequency-component approach as it is based on preprocessing the raw EEG signals using a Power Spectral Density (PSD) analysis followed by a Welch-based smoothing process to reduce the variance and increase the spatial correlation among frequency bins. The latter is crucial for the training and inference of the CNN. The data is normalized and standardized into a coherent electrode placement method, aligned with the international 10-20 system. All the three datasets are down-sampled to a common frequency of 128 Hz, and the number of channels is reduced to 14 as per the number of electrodes in the EEG sensor cap of the lowest resolution. Eye artifacts are removed through Independent Component Analysis (ICA) by comparing the scalp plots, frequency patterns, and the time-independent activation in Event Related Potentials (ERP) of epochs from the same emotion label.

The various experiments, cultural backgrounds and scenarios of the three datasets present a good opportunity to validate our hypothesis about significant frequency contributions under a dominant emotional state. Accordingly, a strong, scary still frame evokes dominant frequency patterns that are similar to a strong, short and scary video. This results in more data from all the three datasets being used to train the generic classifier. Section IV and V describe the conditions under which compatible subjects are involved in training.

III. Classifier Topology

Convolutional Neural networks (CNNs) are inspired by the mammals visual cortex and were originally used for image recognition. In the context of this research, the EEG frequency data is arranged so as to exploit the CNN capabilities in image recognition for recognizing EEG features.

Fig. 1 illustrates the architecture of the valence classifier implemented in TensorFlow [12]. A similar architecture is used for the arousal classifier. The frequency frame is passed to a convolutional layer containing 100 different kernels, whose convolutional results are passed through a rectified linear unit (ReLU), and down-sampled using Maxpooling with a 2×2 pool window. The intermediate activations are flattened to a dense network of three layers (F_{c1}, F_{c2} and F_{c3}). The input, hidden and output layer has 128, 64, and 2 neurons, respectively, with the two output neurons representing the High and Low labels of the emotion metric.

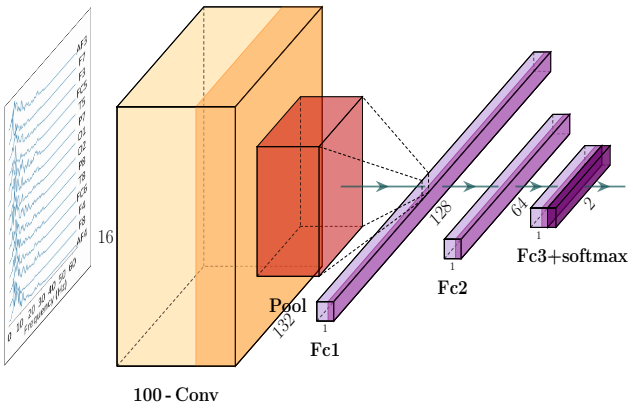

Fig. 1. Architecture of the Valence classifier implemented in Tensorflow

IV. Transfer Learning

Transfer learning is the process of using knowledge gained from a previous situation, to apply it in a new but related context. Assuming a target domain D_t composed by a feature space X_t and a marginal probability distribution $P_t(X_t)$, the predictive function $f_t(.)$ of the target in a learning task T_t could be enhanced by using a source domain D_s [13]. The source domain is composed by a similar pair, $\{X_s, P_s(X_s)\}$, where X_s, P_s, and even the source task T_s satisfy equivalence relationships with their counterpart in the target domain.

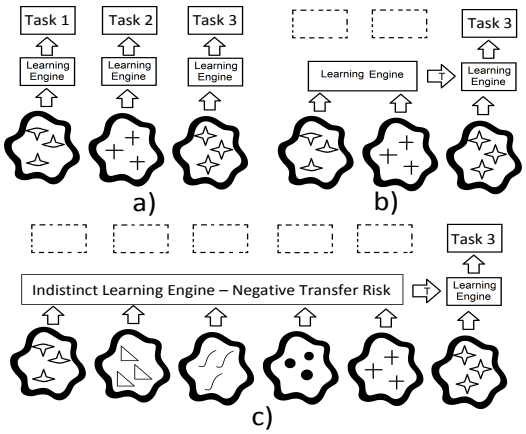

Fig. 2. a) Traditional Machine Learning. b) Unsupervised Transfer Learning. c) Indistinct Transfer Learning with negative transfer risk.

This paper employs *Unsupervised* Transfer learning, which uses various source domains (Training Subjects D_{s1} and D_{s2} in Fig. 2(b) that are related to the target domain (Validation Subject), and whose source and target

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learning tasks (Prediction of emotions) are also different but related. Due to the diverse datasets, the learning tasks (e.g., T_{s1}, T_{s2}, T_{s3}) are not equal but equivalent, with their equivalence generated by selecting experiences evoking the same arousal and valence.

V. Semi-Independent Training

As already mentioned, machine learning (ML) models for emotion detection are traditionally trained using two different techniques: subject-dependent (SD) and subject-independent (SI). Fig. 2(c) illustrates the subject-independent approach, which is prone to a negative transfer effect due to the indistinct training using all subjects. Fig. 2(a) represents the subject-dependent approach. A more precise representation is learned in this subject-specific classifier, but the drawback is the low number of samples. In the *subject semi-independent* (SSI) approach proposed in this paper (Fig. 2(b)), the data from a *subset* of *similar* subjects is used to train the classifier.

A. Two Unsupervised Learning Stages

The ideal training scenario is to cluster subjects with similar EEG frequency patterns under the exact same stimulus. However, since not all the subjects are exposed to the exact same stimulus, it is necessary to introduce a criterion to evaluate equivalence among stimuli. This motivates the use of an unsupervised learning stage to cluster stimuli based on the manual ratings entered by the volunteers. Stimuli that evoke the same arousal and valence are declared to belong to the same zone as defined by the K-means algorithm. This is illustrated in Fig. 3.

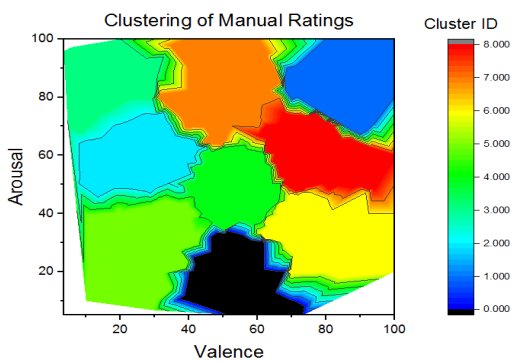

Fig. 3. Manual feedback of all subjects used to create stimulus sections

Once the first unsupervised stage is completed, a second unsupervised learning stage is applied to cluster the EEG responses generated by a set of equivalent stimuli. The K-Means algorithm is now applied to the EEG recordings in a high dimensional space, where the spectrum for each frequency bin of stimuli in the common manual sectors are used as features for each subject. As a result, subjects in the same EEG cluster will end up sharing similar PSD-Welch frequency patterns.

B. Complete Data Flow

Fig. 4 illustrates the complete flow of our data acquisition and ML methodology for a given subject. Two streams of data, manual ratings ($Val\#$) and EEG data, are received in parallel from the experimental setup described in Section II. The manual ratings are allocated in the various zones described in Fig. 3, according to the K-means classifier used in the unsupervised learning stage. The zones where the new subject has stimuli are used to pre-select a subset of subjects for comparison. The parallel branch used to receive the EEG data connects with a pass-band filter (4-45Hz), an ICA module to remove artifacts, and a PSD-Welch module to extract frequency features. Both branches are combined in the EEG clustering stage, whose output is the subset of similar subjects. This subset of similar subjects is used in the off-line, supervised training of the CNN as in Fig. 1, which is further deployed to classify newly acquired test data from the validation subject. The training is achieved using back-propagation and stochastic gradient descent.

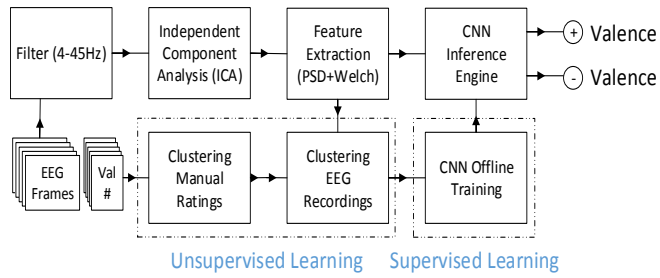

Fig. 4. Block diagram of the Complete Data Flow

VI. Results

The data flow described in Fig. 4 is applied to each subject to obtain their subgroup of volunteers with similar EEG frequency patterns. Fig. 3 illustrates the clustering result for the first unsupervised learning stage based on manual ratings. Only those subjects with stimuli in similar rating zones are exposed to the second clustering stage to assess their EEG similarity. Principal component analysis (PCA) is used in Fig. 5 to present the six general clusters of one subject.

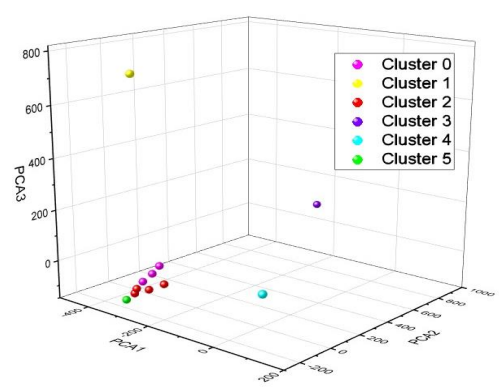

Fig. 5. PCA of the EEG clustering process for subject $sloc01$

Table I displays multiple training alternatives for $sloc01$, combining similar and non-similar subjects intentionally to show the similarity effect. The "x" mark in front of the subjects indicates their inclusion in a training case. For instance,

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TABLE I

SIMILARITY METRICS EXTRACTED FOR A PARTICULAR SUBGROUP

Subject ID	Cluster Label	θ_{sim}	ρ_{sim}	Training Alternatives 1	2	3	4	5	6	7
sdea01	2	1.774	276.830	x		x	x			
sdea24	0	1.585	3528.134	x	x	x				
sdea28	2	1.842	325.007					x		x
sdre14	5	1.520	550.648	x	x		x			
sloc01	2	0	0	x	x	x	x	x	x	x
sloc02	2	1.754	262.486					x	x	
sloc03	1	1.461	861.667							
sloc04	2	1.712	335.904					x	x	x

Alternative 2 includes the class-balanced experimental data of two participants ($sdea24$ and $sdre14$) with high relative distance (ρ_{sim}) to the subject of interest ($sloc01$), which is used as validation subject. Additionally, for each training alternative, 10-fold cross validation (CV) is applied using all subjects ($sdea24$, $sdre14$ and $sloc01$) to measure generalization within the subgroup to which the subject of interest belongs. As expected, the first four training combinations lead to a decaying 10-fold CV accuracy in Fig. 6(b) due to the sparsity implied by Fig. 6(a). The relative direction angle (θ_{sim}) provides insight into the relative vector direction of a subject within the subgroup.

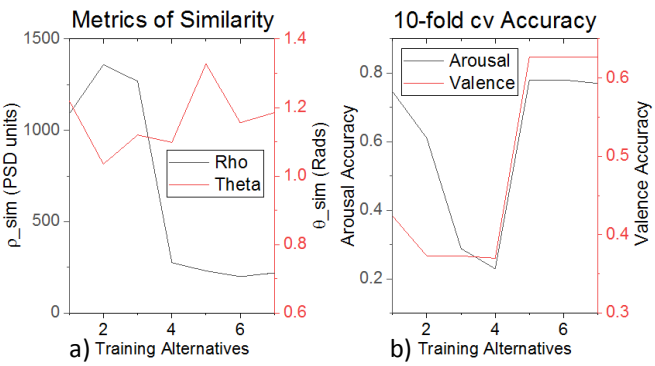

Fig. 6. Metrics of similarity and Arousal/Valence 10-fold CV accuracy

The average accuracies of the state-of-the-art, generic, i.e., subject-independent, classifiers are summarized in Table II. The accuracy is used to compare quantitatively the generalization capabilities of the subject-independent classifiers. The major contribution of this paper is in enabling transfer learning from one subject to another, under well-defined similarity rules. Our accuracy is presented as an average of the individual 10-fold CV results of the reduced cluster associated with each of the 60 subjects. As expected, our subject-semi-independent approach is better than the subject-independent one, but it is not as good as the most complex subject-dependent classifiers.

VII. CONCLUSIONS

The subject semi-independent training for EEG-based emotion detection proposed in this paper comprises rating selection, subject selection, and two unsupervised learning stages. Unsupervised learning is used first to cluster stimuli for data selection and second to cluster similar EEG responses. This results in an increase in the training data of healthy subjects, but potentially patients with ALS or Alzheimer, whose exposure to long experiments is limited. Signal denoising and artifact removal are crucial to avoid biasing the data and subject clustering steps. The results obtained with the CNN imply that the hypothesis made in Section II about keeping the same dominant frequencies for different scenarios of the same evoked experience is valid. We have found out that the usage of frequency features is key for enabling EEG dataset combination for emotion recognition. One important outcome of this work is the standardization and merging of three EEG datasets into a frequency-based repository, named FEXD, which can be used to further refine ML models for emotion detection.

TABLE II

ACCURACY COMPARISON FOR THE STATE-OF-THE-ART

Contribution	Valence	Arousal	Subject	Validation
Mahnob-HCI [4]	57.0	52.4	27	20-fold CV SI
DREAMER [3]	62.49	62.17	23	10-fold CV SI
Our FEXD	**70.26**	**72.42**	**60**	**10-fold CV SSI**

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